Topic 9: Probability and Expectations

(How economists talk about uncertainty, types, and randomization)

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1. Why probability enters now

Up to now, the pre-course has built a toolkit for:

- sets and mappings,
- preferences and utility,
- convexity and geometry,
- continuity, closedness, and existence,
- optimization and correspondences,
- differentiability and marginal reasoning,
- envelope theorems and value functions,
- order, lattices, and monotone comparative statics.

All of these tools were developed in a world of certainty. The agent knows her choice set, she knows her utility function, and she knows the parameters of the problem. The only question was: given all this information, what does she choose, and how does that choice change when the environment changes?

But a large part of microeconomics does not live in such a world.

Agents often face:

- uncertainty about payoffs (will this investment pay off?),
- uncertainty about other agents' characteristics (what is the other bidder's valuation?),
- uncertainty about signals (what does this test result mean?),
- or deliberate randomization (a player who mixes between strategies).

So before we can talk cleanly about auctions, Bayesian games, or mixed strategies, we need a minimal probability language.

The point of this topic is not to turn you into a probability theorist. It is to make sure that when economists write "expected payoff" or "distribution over types," those words are no longer mysterious.

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2. The state space: what might happen

Start with the most primitive object: the set of things that could happen.

We call this set Ω and refer to it as the **state space**. Each element $\omega \in \Omega$ is a **state of the world** – a complete description of everything that is uncertain.

In economics, the state space might describe:

- which of several possible cost realizations occurs,
- which type each player in a game has,
- which signal each agent observes.

> "Ω is the set of all possible states of the world. An element ω is one particular resolution of uncertainty."

Mathematically, this is just a set like any other. It becomes interesting only once we start measuring how likely different parts of it are.

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3. Events and probabilities

An **event** is a subset of the state space: $A \subseteq \Omega$.

> "An event is a collection of states. The event A occurs if the realized state ω happens to lie in A."

A **probability** is a rule that assigns a number to events in a coherent way. Formally, a probability (often written Pr or P) satisfies three properties:

1. **Nonnegativity**: for every event A,

$$\Pr(A) \geq 0.$$

2. **Normalization**: the probability that *something* happens is one,

$$\Pr(\Omega) = 1.$$

3. **Additivity**: if A and B are disjoint events (they share no states), then

$$\Pr(A \cup B) = \Pr(A) + \Pr(B).$$

> "Probability is a coherent way of assigning weights to events. The weights are nonnegative, the total is one, and non-overlapping events add up."

These three properties are sometimes called the **Kolmogorov axioms**. They are the foundation on which everything else rests.

Note that we said nothing about *where* probabilities come from. In some models they represent objective frequencies (how often something happens if we repeat the experiment). In others they represent subjective beliefs (how confident the agent is). The mathematics is the same either way. Microeconomics uses both interpretations depending on the context.

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4. Random variables

A **random variable** is a mapping from states of the world into numbers:

$$X : \Omega \to \mathbb{R}.$$

> "A random variable turns uncertainty about states into uncertainty about a number."

This is just a function in the sense of Topic 1. The domain is the state space Ω, the codomain is $\mathbb{R}$. What makes it "random" is that we do not know which ω will occur, and therefore we do not know what value $X(\omega)$ will take.

In economics, random variables represent objects like:

- a consumer's income (uncertain because it depends on the state of the economy),
- a bidder's valuation in an auction (uncertain because it is private information),
- a firm's cost (uncertain because of technological shocks),
- a player's type in a game of incomplete information.

Why does this matter? Because once uncertainty is encoded as a random variable, we can use all the tools of analysis – integration, optimization, comparison – to study it systematically.

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5. Distributions

Knowing that X is a random variable is not enough. We also need to know *how likely* different values of X are. This is what a **distribution** tells us.

Discrete distributions

If X takes finitely many values $x_1, \ldots, x_n$, the distribution is simply a list of probabilities:

$$p_i = \Pr(X = x_i), \quad i = 1, \ldots, n.$$

The probabilities are nonnegative and sum to one:

$$\sum_{i=1}^{n} p_i = 1.$$

This is the simplest case. It covers situations like: a firm faces two possible cost levels, or a buyer's type is drawn from a finite set.

Continuous distributions and the CDF

In many economic models, the random variable takes a continuum of values. We do that because it makes our lives often much much easier. In that case, the probability that X equals any single value is typically zero, so we need a different way to describe the distribution.

The **cumulative distribution function** (CDF) is defined as

$$F(x) = \Pr(X \le x).$$

> "The CDF at x is the probability that the random variable takes a value no larger than x."

Properties of the CDF:

- F is nondecreasing (higher x means weakly more probability accumulated),
- $\lim_{x\to-\infty} F(x) = 0$ and $\lim_{x\to+\infty} F(x) = 1$,
- F is right-continuous.

If the CDF is differentiable, we can define the **probability density function** (PDF) as

$$f(x) = F'(x).$$

The density tells us how "concentrated" probability is near x. A higher density means more probability mass near that value.

Economists often write things like "let v be drawn from F" or "types are distributed according to F." This simply means: v is a random variable with cumulative distribution function F.

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6. Expectation: the probability-weighted average

The **expectation** (or **expected value**) of a random variable is its probability-weighted average.

For a discrete random variable with values $x_1, \ldots, x_n$ and probabilities $p_1, \ldots, p_n$,

$$\mathbb{E}[X] = \sum_{i=1}^{n} p_i\, x_i.$$

For a continuous random variable with density f,[1]

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x\, f(x)\, \mathrm{d}x.$$

[1] I usually claim that integrals $\int$ are nothing but a fast way of writing a sum $\sum$. Although mathematically there is a difference and the laws are a bit different, this intuition is often good for *economic* purposes. In reality people have a hard time thinking in terms of continuous state spaces, so even if they were they behave as if things were discrete. If you result relies on one of the tricks where the integral behaves different from the sum, there may be an issue in what you are modeling.

> "Expectation is the weighted average value you would get if the uncertainty were resolved many times."

Why economists care about expectation

Expectation is the central bridge between probability and optimization. Whenever an agent faces uncertainty and must choose *before* the uncertainty resolves, the standard approach in economics is to evaluate choices by their **expected payoff**.

This appears as:

- **expected utility** in consumer theory under risk,
- **expected profit** in a firm's production decision,
- **expected revenue** in auction design,
- **expected continuation payoff** in dynamic games.

Linearity of expectation

One of the most useful properties of expectation is **linearity**:

$$\mathbb{E}[aX + bY] = a\,\mathbb{E}[X] + b\,\mathbb{E}[Y]$$

for any constants a, b and random variables X, Y.

> "The expected value of a sum is the sum of the expected values, regardless of whether the random variables are related."

This property holds even when X and Y are not independent. It is one of the reasons expectation is so tractable.

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7. Conditional probability and Bayes' rule

Sometimes new information arrives. A bidder observes a signal. A firm learns a partial cost estimate. An agent sees another agent's action.

When information arrives, beliefs must be updated. This is where **conditional probability** enters.

Conditional probability

The probability of an event A conditional on observing that event B has occurred is

$$\Pr(A \mid B) = \frac{\Pr(A \cap B)}{\Pr(B)},$$

provided $\Pr(B) > 0$.

In English the above equation says:

> "The conditional probability of A given B is the probability that both A and B occur, divided by the probability that B occurs."

Think of it as zooming in on the world where B happened and asking: within that world, how likely is A?

Bayes' rule

Bayes' rule is a way to represent the previous equation as a relation between conditional probabilities. It thus follow directly from the above rule

$$\Pr(A \mid B) = \frac{\Pr(B \mid A)\,\Pr(A)}{\Pr(B)}.$$

> "Bayes' rule says that when we observe B, the probability of A is the probability that B has occurred *because* A is true, times the relative liklihod of A occurring in the first place, all divided by the overall probability of B."

This formula is the formal engine behind belief updating in microeconomics. The components have names:

- $\Pr(A)$ is the **prior** – what we believed before seeing B,
- $\Pr(B \mid A)$ is the **likelihood** – how probable the evidence B is if A were true,
- $\Pr(A \mid B)$ is the **posterior** – what we believe after seeing B.

Bayes' rule is used everywhere in information economics. Whenever you read "the agent updates beliefs," the formal content behind that phrase is Bayes' rule.

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8. Independence

Two events A and B are **independent** if

$$\Pr(A \cap B) = \Pr(A) \cdot \Pr(B).$$

So the probability that both A and B occur is just the product of their individual probabilities.

Two random variables X and Y are **independent** if learning the value of one tells you nothing about the other. Formally, this means that for all values x and y,

$$\Pr(X \leq x \text{ and } Y \leq y) = \Pr(X \leq x) \cdot \Pr(Y \leq y).$$

> "Independence means that observing one thing does not change your beliefs about the other."

Independence is a simplifying assumption that appears constantly in economic models. For instance:

- in auctions, bidders' valuations are often assumed to be independently drawn,
- in mechanism design, agents' types are frequently independent,
- in repeated games, stage-game shocks may be independent across periods.

Important: Independence is substantive. It is not a default. In reality, everything is connected with everything and many economically interesting situations involve correlation (e.g., common-value auctions, correlated types). When independence is assumed, it should be recognized as a modeling choice, not a law of nature.

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9. Conditional expectation

When we combine expectation with conditional probability, we get **conditional expectation**:

$$\mathbb{E}[X \mid B] = \sum_i x_i \operatorname{Pr}(X = x_i \mid B)$$

in the discrete case.

> "The conditional expectation of X given B is the expected value of X knowing that B is true."

In economics, conditional expectations appear naturally whenever agents form beliefs:

- a buyer's expected valuation given a signal,
- a principal's expected output given the agent's reported type,
- a player's expected payoff given the other players' strategies.

A useful property that connects conditional and unconditional expectation is the **law of iterated expectations** (sometimes called the law of total expectation):

$$\mathbb{E}[X] = \mathbb{E}\big[\mathbb{E}[X \mid Y]\big].$$

> "If I first compute the expected value of X for each possible value of Y, and then average those conditional expectations over the distribution of Y, I recover the unconditional expectation of X."

This property is extremely useful in sequential models where information is revealed gradually.

It is also super important for consistency and tells us one way to think about distributions (that will show up in other settings, too). We could think of a larger sate space Ω that has two dimensional elements (x, y) and a probability distribution over those (x, y). Now, we are in the end only interested in x, but y carries some information about x.

So, if we learn y, we update about x. But: it has to be the case that if we don't know y yet, we cannot learn anything about x, so our expectations on x are just

$E[X]$. But, we do know how likely all thedifferent y will show up and we know what we would do then in our beliefs about x, so we do know all $E[X|Y]$. But if we are consistent, then this information does not tell us anything about x yet (because we haven't seen y yet), so we have to have that $E[X] = E[E[X|Y]]$.

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10. An economic example: first-price auction with private values

Consider two bidders in a sealed-bid first-price auction. Each bidder has a private valuation v_i drawn independently from a distribution F on $[0, 1]$. Each bidder submits a bid b_i, the highest bid wins, and the winner pays her bid.

Bidder 1's expected payoff from bidding b when her valuation is v and the other bidder uses a bidding strategy $\beta(\cdot)$ is

$$\mathbb{E}[\text{payoff}] = (v - b) \cdot \Pr(\text{win}).$$

The probability of winning depends on the other bidder's bid, which depends on her valuation, which is random. So

$$\Pr(\text{win}) = \Pr(\beta(v_2) < b) = \Pr(v_2 < \beta^{-1}(b)) = F(\beta^{-1}(b)),$$

where the second equality uses the fact that β is (typically) increasing.

This example uses almost everything from this topic:

- a random variable (v_2, the other bidder's type),
- a distribution (F),
- expectation (the expected payoff),
- independence (valuations are independently drawn).

Without the probability language, writing down and solving this problem would be impossible.

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11. Where this appears in Mas-Colell, Whinston, and Green (MWG)

Probability is used throughout MWG but is not developed as a standalone mathematical appendix topic in the same way as convexity or correspondences.

The most relevant places where probability language is used heavily include:

- **MWG, Chapter 6** (choice under uncertainty, expected utility),
- **MWG, Chapter 8** (competitive markets with uncertainty),
- **MWG, Chapter 13** (adverse selection),
- **MWG, Chapter 23** (mechanism design and auctions).

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12. Why this matters for microeconomics

Probability gives us the language to treat uncertainty as part of the model rather than as narrative.

Without it:

- expected utility cannot be defined,
- Bayesian equilibrium cannot be formulated,
- mixed strategies appear as magic rather than precise objects,
- information economics loses its formal language,
- mechanism design under incomplete information becomes inaccessible.

The tools in this topic are modest – we have not touched measure theory, stochastic processes, or advanced convergence results. But they are exactly what is needed to read the core microeconomics curriculum without stumbling over probabilistic language. And they already get us a long way into the heart of modern microeconomic theory.

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13. What you should take away

After this topic, you should be comfortable with:

- reading statements about random variables and distributions,
- computing and interpreting expectations as weighted averages,
- using conditional probability and Bayes' rule to update beliefs,
- recognizing independence as a substantive modeling assumption,
- and seeing why uncertainty is unavoidable in modern microeconomics.

You should also see how this topic connects back to earlier material: random variables are functions (Topic 1), distributions live on sets (Topic 1), expected utility combines preferences (Topic 2) with probability, and optimization under uncertainty (Topics 4 and 5) evaluates choices by expected payoffs.